Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Jeff Coller	Email	tipa@jhu.edu
Institute Name	Johns Hopkins University			
Address	725 N. Wolfe St. Baltimore MD 21205			

2. Technology Overview

Title of Proposed Technology	Advanced Nanocarrier Platforms for Multimodal and Combination Therapeutic Delivery for base editing and RNA-based genome modulation			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio/ Medical		Bio/ Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Drug Delivery Systems	Medical Bio	Biomaterials
Concept of Proposed Technology	Many emerging therapies require simultaneous or coordinated delivery of therapeutics with different physicochemical properties, including small molecules, proteins, and mRNA-based genome-modulating payloads, yet current delivery technologies often exhibit limited payload compatibility, suboptimal intracellular delivery, off-target accumulation, and immune-related toxicity *[1,2]*. This joint R&D program seeks proposals focused on nanocarrier platforms that enable efficient intracellular delivery of diverse therapeutic payloads, with emphasis on: • High-capacity and stable loading of both hydrophilic and hydrophobic agents, or multiple biologically distinct payloads; • Direct or efficient cytosolic delivery mechanisms that reduce reliance on endosomal escape *[4]*; • Biocompatible and biodegradable materials minimizing immune activation and systemic toxicity; • Modular surface engineering enabling tissue- or cell-type-specific targeting; • Compatibility with scalable and quality-controlled manufacturing processes.			

(max 10 pages)

	The program does not prescribe specific nanocarrier geometries, materials, or targeting ligands, emphasizing instead functional delivery performance, safety, and translational readiness.
Rationale for Joint R&D Project	Clinically successful drug delivery platforms require coordinated progress in carrier design, biological mechanism validation, formulation robustness, and process scale-up. • U.S. research institutions contribute strengths in RNA biology, mRNA engineering, base editing mechanisms, and intracellular trafficking. • Korean startups/SMEs contribute strengths in formulation optimization, analytical validation, manufacturing process development, and regulatory preparation. A Korea–US joint R&D framework enables efficient maturation of nanocarrier technologies from laboratory discovery to commercialization-ready delivery platforms, while mitigating scientific and regulatory risks.
Key Words	1. Drug delivery system (DDS) 2. Nucleic acid combination therapeutics 3. Lipid-based particles 4. Targeted and intracellular delivery

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4)	**End of the Project**	Level (6)
R&D Objectives	Proposals should aim to: 1. Develop nanocarrier formulations capable of delivering multiple or diverse therapeutic payloads *[3]*; 2. Demonstrate efficient intracellular delivery and functional payload activity; 3. Establish robust and scalable formulation and manufacturing processes; 4. Validate safety, stability, and targeting feasibility in preclinical settings; 5. Demonstrate application-specific proof-of-concept supporting downstream development. 6. Demonstrate functional genome editing activity mediated by delivered base-editing mRNA. Projects are expected to progress from laboratory feasibility (TRL 3–4) toward prototype and pilot-scale validation (TRL 5–6).			
R&D Target Specifications	1. Multimodal Payload Compatibility • Demonstrate stable encapsulation and co-delivery capability for chemically and biologically distinct therapeutic agents. 2. Intracellular Delivery Efficiency • Establish evidence of effective cytosolic delivery and functional payload activity in relevant cell models. 3. Formulation Stability and Safety • Maintain physicochemical stability and acceptable cytotoxicity under biologically relevant conditions.			

(max 10 pages)

	4. Targeting and Biodistribution Feasibility • Demonstrate potential for tissue- or cell-selective delivery through surface modification or formulation design. 5. Scalable Manufacturing Readiness • Establish reproducible and quality-controlled formulation and production processes suitable for pilot-scale manufacturing. 6. Application-Specific Proof-of-Concept • Demonstrate therapeutic relevance in at least one representative disease or modality setting.
R&D Plan	**YEAR 1 — Formulation Feasibility and Payload Delivery Validation (TRL 3–4)** Goal: Establish feasibility of multimodal payload loading and intracellular delivery. • Design and screen nanocarrier formulations for diverse payload compatibility. • Evaluate intracellular delivery mechanisms and functional payload expression or activity *[5]*. • Perform physicochemical and biological stability assessments. • Conduct *in vitro* safety and preliminary targeting evaluations. • Select candidate formulations for further development. Indicative Deliverables: • Feasible multimodal delivery formulations • Evidence of intracellular functional delivery • Preliminary safety and stability profiles **YEAR 2 — Process Optimization and Preclinical Proof-of-Concept (TRL 4–5)** Goal: Demonstrate therapeutic relevance and manufacturing feasibility. • Optimize formulation parameters for stability, delivery efficiency, and reproducibility. • Establish manufacturing-aligned production and quality attributes. • Demonstrate application-specific proof-of-concept in relevant disease models. • Evaluate targeting strategies and biodistribution characteristics. • Refine analytical and quality control methods. Indicative Deliverables: • Optimized and scalable formulation process • Preclinical efficacy and targeting evidence • Defined critical quality attributes **YEAR 3 — Reliability, Scale-Up, and Translation Preparation (TRL 5–6)** Goal: Establish robustness and readiness for downstream development. • Demonstrate batch-to-batch reproducibility and long-term stability. • Conduct extended safety and compatibility assessments. • Finalize pilot-scale manufacturing workflows and QC frameworks. • Prepare regulatory-supporting technical documentation. • Establish commercialization and partnership strategies. Indicative Deliverables:

(max 10 pages)

	• Pilot-scale validated delivery platform • Regulatory-ready preclinical data package • Commercialization and licensing roadmap

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Payload design and biological activity assessment • mRNA engineering and stability optimization • Base editing efficiency and mechanism-of-action validation
Expected R&R of Korean SME Partner	• Nanocarrier formulation and optimization • Process scale-up and manufacturing feasibility • Quality systems, regulatory preparation, and commercialization planning

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• Versatile drug delivery platforms enabling new combination and nucleic-acid therapies *[6]* • Reduced dependence on restricted or heavily patented delivery technologies • Accelerated global competitiveness of Korean biopharma SMEs • Strengthened Korea–US collaboration in advanced drug delivery innovation

6. References

(A) Limitations of Lipid Nanoparticles (LNPs) and Conventional Delivery Systems

1. Tenchov, R. et al., Lipid nanoparticles from liposomes to mRNA vaccines, ACS Nano, 2021.

2. Hou, X. et al., Lipid nanoparticles for mRNA delivery, Nature Reviews Materials, 2021.

(B) Need for Multimodal and Combination Therapeutic Delivery Platforms

3. Li, X. et al., Nanoparticle technologies for mRNA delivery: strategies, challenges, and opportunities, Frontiers in Bioengineering and Biotechnology, 2024.

4. Blanco, E. et al., Principles of nanoparticle design for overcoming biological barriers , Nature Biotechnology, 2015.

(C) Cytosolic Delivery and Next-Generation Nanocarrier Platforms

5. Kowalski, P. S. et al., Delivering the messenger: advances in technologies for therapeutic mRNA delivery., Molecular Therapy, 2019.

6. Mitchell, M. J. et al., Engineering precision nanoparticles for drug delivery, Nature Reviews Drug Discovery, 2021.

Recent studies highlight critical limitations of conventional lipid-based delivery systems, particularly in multimodal payload compatibility, cytosolic delivery efficiency, and tissue specificity, underscoring the need for next-generation nanocarrier platforms.